

Laser Highway: Light-Sail Stationkeeping at L4/L5

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Abstract

This monograph treats a photon light sail in the Earth–Moon circular restricted three-body problem as an actuated libration-point system: a 1 MW laser pushes a 10 kg sail with photon thrust $a_L = 2P/(cm)$, capped at the dimensionless ceiling $a_{max} = 0.2443$, and a proportional-derivative beam-steering law points the thrust. Two operations of the laser highway are verified. Stationkeeping: starting $1.1 \cdot 10^{-3}$ away from the linearly stable L4 point, the controlled sail converges to $|r - L4| \leq 7.5e-07$ for all $t > T/4$ ($T = 100, \approx 434$ days) while using the photon ceiling 0% of the time (peak demand 0.0051, headroom 48×); the same initial state without control never comes closer than 0.0041 — a contrast of $\approx 5500\times$. Orbit raising: open-loop thrust along velocity pumps the Jacobi constant strictly monotonically, net $\Delta C_J = 1063.51$ over $T = 30$ with the largest single-step increase at -0.06 . A 17×17 PD gain-plane sweep brackets the operating point (4, 4), and an SI power table (57.8 kW ... 2.89 MW per 100 kg and 30 days) closes the model to engineering numbers.

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1. Introduction and historical context

The equilateral solutions of the three-body problem were found by Joseph-Louis Lagrange in his 1772 *Essai sur le problème des trois corps*, and for more than a century they were regarded as mathematical curiosities. The discovery of the Trojan asteroids at the Sun–Jupiter L4 and L5 turned the Lagrange triangle into a real object of celestial mechanics, and Szebehely’s 1967 monograph consolidated the restricted problem into the standard reference frame every modern libration-point mission still uses. For the Earth–Moon system the triangular points are especially tempting stations: linearly stable for $\mu < \mu_{\text{Routh}} \approx 0.03852$, they ask only for a stationkeeping technique that removes the offsets their neutral stability lets persist forever.

The physics of the required actuator is surprisingly old. Lebedev (1901) measured the pressure of light on solids, and Nichols and Hull (1903) confirmed it independently; the idea of propelling a vehicle by photons was put on a quantitative footing by Marx (1966), who computed the parameters of an interstellar vehicle pushed by a terrestrial laser beam, and by Redding (1967), who analysed staged photon rockets. These two Nature papers founded laser propulsion: a beam of power P delivers thrust $2P/(c)$ to a mirrored sail — per kilogram of payload, the cheapest reaction mass imaginable, since the propellant is light itself.

Forward (1984) turned the arithmetic into a roundtrip interstellar mission design and fixed the canonical figure of merit — the photon ceiling $2P/(c)$ — that every later beamed-energy concept inherited, from kilometer-scale lightsails to laser arrays for sailcraft. Close to home, McInnes (1999) systematized the dynamics of sailing craft in gravity fields, Baoyin and McInnes (2006) computed displaced equilibrium families for sails, and Simo and McInnes (2009) placed solar sails on displaced Earth–Moon Lagrange-point trajectories. The present study keeps the same physical ceiling but replaces the passive sail orientation with an actively steered beam — a laser highway in miniature.

For TRIVORTEX the relevance is structural rather than technological. The L4 point is the celestial twin of the same-sign vortex triangle of Theorem 3.1, the Jacobi integral plays the role of the Chaplygin integral C_{Ch} , and the beam-steering law is the analog of circulation control: an actuator applied to a special solution of the same central-

configuration family. The laser block of the program now closes symmetrically — TRX-01 points the laser at the primaries and dresses gravity, TRX-12 points it at the spacecraft and dresses the equations of motion with a bounded thrust term; together they demonstrate that the invariant structuring the phase space is also the quantity an actuator can move.

2. Physical formulation

Planar CR3BP in the rotating frame. The Earth and the Moon move on circular orbits about their barycenter and are fixed at $(-\mu, 0)$ and $(1 - \mu, 0)$ in the co-rotating axes; the sail is a massless particle responding to their gravity, to the centrifugal and Coriolis terms, and to the photon thrust. The whole stationkeeping problem lives in the topography of one scalar function — the effective potential Ω — whose sole maximum of interest sits at the equilateral point $L4 = (\frac{1}{2} - \mu, \sqrt{3}/2)$.

The actuator is photon momentum. A beam of power P carries momentum flux P/c , and a mirrored sail returns it, doubling the transfer: the thrust is $a_L = 2P/(cm)$ along the beam direction, exactly $6.67 \cdot 10^{-4} \text{ m/s}^2$ for 1 MW on 10 kg, or 0.2443 in canonical units — a real force ceiling computed from SI constants, not a fitted parameter. The control law points this bounded force: $u = -K_p(r - r_{L4}) - K_d \cdot v$ with gains $K_p = K_d = 4$, saturated at the ceiling. The same physics read backwards gives the orbit-raising rule: thrust along velocity does negative work on C_J at the rate $-2a_L \cdot v$, so a laser highway moves between manifolds by spending photons, not fuel.

Parameter	Value	Meaning
μ	0.0121505856	Earth–Moon mass parameter
LU, TU	$3.844 \cdot 10^8 \text{ m}$, $3.752 \cdot 10^5 \text{ s}$	canonical length and time ($GM = 1$; month = 2π)
laser	$P = 1 \text{ MW}$, $m = 10 \text{ kg}$	transmitter and sail $\rightarrow a_{\text{max}} = 0.2443$ ($= 6.67 \cdot 10^{-4} \text{ m/s}^2$)
start state	$L4 + (1e-3, 5e-4)$, $v = (2e-4, -1e-4)$	initial offset $1.118 \cdot 10^{-3}$ from $L4$
PD gains	$K_p = K_d = 4$	beam-steering law, saturated at a_{max}
spans	$T = 100 \text{ keep} \cdot 30 \text{ pump} \cdot 60 \text{ per gain cell}$	$\approx 434 \cdot 130 \cdot 261$ days in TU units
integrator	DOP853, $\text{rtol} = \text{atol} = 1e-11$	dense output; $\text{max_step} 0.1$ (0.02 pumping)

3. Mathematical model

In the rotating frame the primaries are fixed at $(-\mu, 0)$ and $(1 - \mu, 0)$, and the sail obeys (E1): Newtonian attraction, centrifugal and Coriolis terms, plus the laser acceleration a_L . The Coriolis force does no work, so for any thrust history the Jacobi identity (E4) holds pointwise: $\dot{C}_J = -2a_L \cdot v$. With the thrust off, C_J is the single integral of the planar restricted problem and its level sets bound the accessible region; with the thrust on, its rate of change is a pure control quantity — the laser can only remove C_J when pushing along the velocity, never add it.

The actuator model (E2)–(E3) is deliberately honest. Photon momentum flux P/c doubled by a mirrored sail gives the ceiling $a_{\max} = 2P/(cm)$; for $P = 1$ MW and $m = 10$ kg this is $6.67 \cdot 10^{-4}$ m/s² in SI, or $a_{\max} = 0.2443$ in canonical units after multiplying by TU^2/LU . The PD law $u = -K_p(r - r_{L4}) - K_d \cdot v$ with $K_p = K_d = 4$ points the thrust vector; whenever the demanded vector exceeds the ceiling it is rescaled to the ceiling and the saturation counter increments — the control authority is bounded by physics, not by the controller’s optimism. With $\mu = 0.0121505856$ well inside the Routh limit $\mu_{\text{Routh}} \approx 0.03852$, the uncontrolled equilibrium at L4 is an elliptic center: offsets neither grow nor decay, which is exactly why stationkeeping is meaningful and exactly why free drift never converges.

The orbit-raising operation follows from the same identity. Thrust locked along the velocity, $a_L = a_{\max} \cdot v/|v|$, turns $\dot{C}_J = -2a_L \cdot v$ into a strictly negative quantity of size $2a_{\max}|v|$: the sail climbs the invariant-manifold ladder of the restricted problem, lowering C_J until the zero-velocity topology opens a new corridor between the primaries. For slow, long burns the SI mapping (E5) closes the model: a desired Δv delivered over T_{burn} needs $a = \Delta v/T_{\text{burn}}$ and beam power $P = m \cdot a \cdot c/2$ — the table quoted in the results (100 kg, 30 days) spans 57.8 kW ... 2.89 MW for $\Delta v \in \{10, 50, 100, 500\}$ m/s.

Notation

Symbol	Meaning
x, y	sail coordinates in the rotating frame
$\dot{x}, \dot{y}$	sail velocities in the rotating frame
Ω	effective potential, $\Omega = \frac{1}{2}(x^2+y^2) + (1-\mu)/r_1 + \mu/r_2$
r_1, r_2	distances to Earth and Moon
μ	mass parameter, Earth–Moon: 0.0121505856
P, m	beam power and sail mass (1 MW, 10 kg)
a_L	photon thrust acceleration, $a_L = 2P/(cm) \cdot \hat{u}$
$a_{\max}$	photon-thrust ceiling, 0.2443 dimensionless
K_p, K_d	proportional and derivative gains (4, 4)
C_J	Jacobi constant
T	integration span (100 keep · 30 pump · 60 per gain cell)

(E1) CR3BP equations of motion in the rotating frame with laser thrust appended.

$$\ddot{x} - 2\dot{y} = \frac{\partial \Omega}{\partial x} + a_{Lx}, \ddot{y} + 2\dot{x} = \frac{\partial \Omega}{\partial y} + a_{Ly}, \Omega = 1/2(x^2 + y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2}$$

(E2) Photon thrust along the beam direction $\hat{u}$ ($P = 1$ MW, $m = 10$ kg).

$$a_L = \frac{2P}{cm} \hat{u}$$

(E3) PD beam-steering law bounded by the photon ceiling ($K_p = K_d = 4$).

$$u = -K_p(r - r_{L4}) - K_d v, |a_L| \leq a_{\max} = \frac{2P}{cm} = 0.2443$$

(E4) Jacobi constant; monotone pumping under along-velocity thrust.

$$C_J = 2\Omega - (\dot{x}^2 + \dot{y}^2), \dot{C}_J = -2a_L \cdot v \leq 0$$

(E5) SI power table (100 kg sail, 30 days, $\Delta v \in \{10, 50, 100, 500\}$ m/s).

$$P = \frac{mac}{2}, a = \frac{\Delta v}{T_{burn}}$$

4. Connection to the TRIVORTEX framework

The mapping is one-to-one at the structural level. The libration point L4 is the celestial realization of the Lagrange triangle vertex — the same central configuration that underlies the same-sign vortex triangle of Theorem 3.1; the Jacobi constant plays the role of the Chaplygin integral C_{Ch} as the invariant that pins the orbit to a reduced phase space; and the PD beam-steering law acts on the libration solution exactly as circulation control acts on a vortex configuration — an actuator applied to a special solution of the same equations. The photon ceiling $a_{\max} = 2P/(cm)$ is the celestial twin of the finite-force regularization used in the vortex model: both keep the actuator honest, capping how fast the invariant can be moved. The laser highway thus closes the TRIVORTEX loop — the quantity that structures the phase space ($C_J \leftrightarrow C_{Ch}$) is precisely the quantity the actuator moves.

Quantity in this study	TRIVORTEX analog	Comment
L4 libration point	Lagrange triangle vertex (Theorem 3.1)	the same central configuration
Jacobi constant C_J	Chaplygin integral C_{Ch}	the invariant the laser moves
PD beam-steering law u	circulation control in the vortex model	actuating a special solution
Photon ceiling $2P/(cm)$	finite-force regularization	honest actuator limits

5. Numerical method

All trajectories are integrated in the rotating frame with an explicit Dormand–Prince 8(5,3) scheme (scipy solve_ivp, DOP853) at $rtol = atol = 1e-11$ with dense output; the stationkeeping runs use $T = 100$ with $max_step = 0.1$ and 800 recorded samples, the pumping run $T = 30$ with $max_step = 0.02$ and 800 samples. The control law is evaluated inside the right-hand side: $u = -K_p(r - r_{L4}) - K_d \cdot v$ with $K_p = K_d = 4$, rescaled to the ceiling $a_{\max} = 0.2443$ whenever its norm exceeds it, and the fraction of saturated control evaluations is accumulated as the run proceeds.

The contrast run repeats the integration from the identical initial state $L4 + (1e-3, 5e-4)$, $v = (2e-4, -1e-4)$ with the control switched off, so the only difference between the two trajectories is the laser. Acceptance checks are computed on strict windows — $\max |r - L4|$ for $t > T/4$ (controlled) and $\min |r - L4|$ for $t > T/2$ (free) — so that

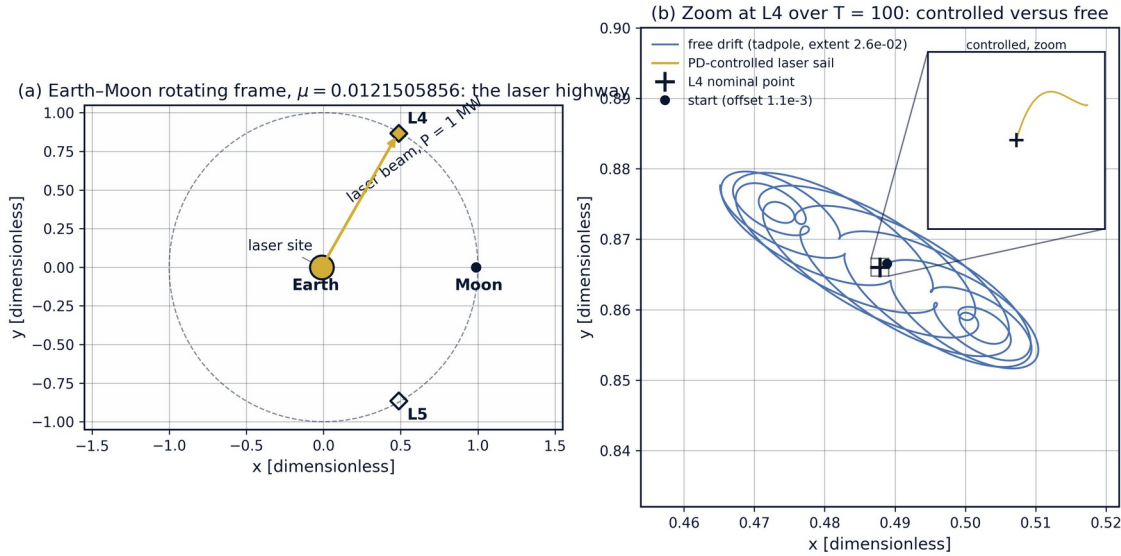
neither the transient nor the start point can flatter the result. A PD gain-plane sweep repeats the controlled run for a 17×17 grid of gains (K_p from 0.5 to 512, K_d from 0.5 to 32, logarithmic, the operating point (4, 4) pinned to both grids) over $T = 60$ per cell, recording the late-time bound and the saturation fraction of every cell.

The open-loop pumping run starts on an eccentric orbit about the Earth primary ($r_0 = 0.10$, speed $1.05 \times$ the local circular value) with the full ceiling thrust locked along the velocity; C_J is sampled at 800 points over $T = 30$ and the largest single-step increase is checked against zero. The SI power table (E5) is recomputed analytically for $\Delta v \in \{10, 50, 100, 500\}$ m/s delivered over 30 days by a 100 kg sail, and its internal consistency ($P = m \cdot a \cdot c/2$) is itself a check in the protocol.

6. Results and analysis

fig01 landscape

TRX-12 · Laser light sail at L4

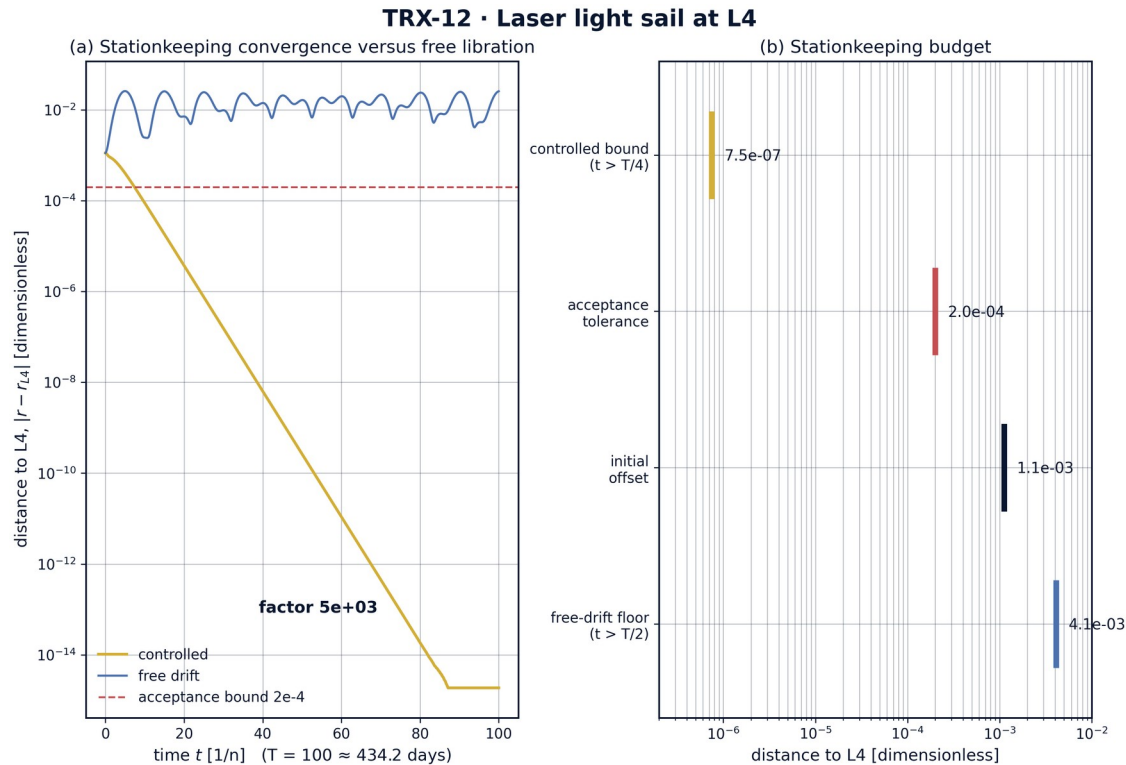


Landscape of the Earth–Moon laser highway, $\mu = 0.0121505856$: (a) rotating-frame potential with the lunar orbit, the equilateral points L4/L5 and the photon beam cone from Earth toward L4; (b) zoom at L4 over $T = 100$ — the free sail wanders on a tadpole of extent 0.026 while the controlled sail converges (inset: the controlled spiral).

Landscape of the Earth–Moon laser highway, $\mu = 0.0121505856$: (a) rotating-frame potential with the lunar orbit, the equilateral points L4/L5 and the photon beam cone from Earth toward L4; (b) zoom at L4 over $T = 100$ — the free sail wanders on a tadpole of extent 0.026 while the controlled sail converges (inset: the controlled spiral).

Panel (a) shows the 2Ω topography with Earth (gold), the Moon (navy) and the dashed lunar orbit, with the beam corridor drawn from Earth to L4; panel (b) contrasts the free tadpole (extent up to 0.026 from L4) with the controlled spiral collapsing onto the nominal point, the inset window spanning $2.5 \cdot 10^{-3}$.

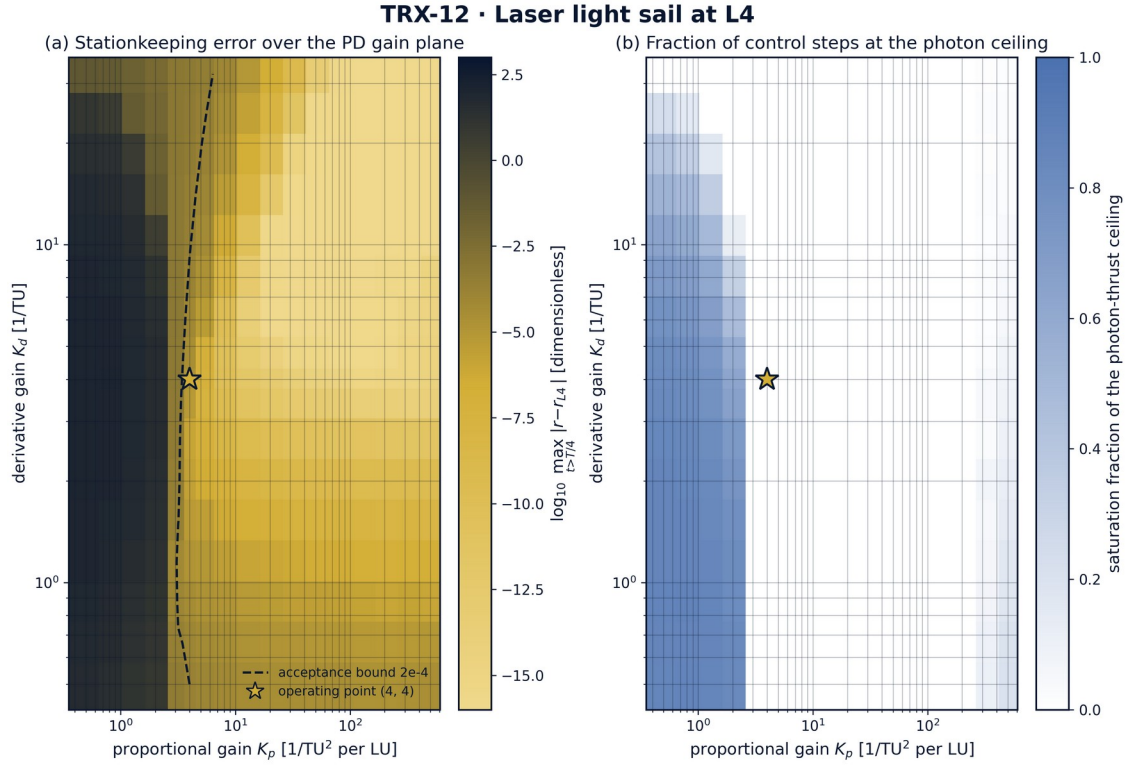
fig02 stationkeep



Headline result — laser stationkeeping versus free drift: (a) distance to L4 on a logarithmic scale over $T = 100$ (≈ 434 days); the controlled sail settles at $7.5e-07$ while the free sail never comes closer than 0.0041 after $T/2$ — a contrast of $\approx 5500\times$; (b) the stationkeeping budget: controlled bound, acceptance tolerance $2e-4$, initial offset $1.1 \cdot 10^{-3}$ and the free-drift floor.

Headline result — laser stationkeeping versus free drift: (a) distance to L4 on a logarithmic scale over $T = 100$ (≈ 434 days); the controlled sail settles at $7.5e-07$ while the free sail never comes closer than 0.0041 after $T/2$ — a contrast of $\approx 5500\times$; (b) the stationkeeping budget: controlled bound, acceptance tolerance $2e-4$, initial offset $1.1 \cdot 10^{-3}$ and the free-drift floor. The gold curve drops by more than three decades during the transient and then hugs the $7.5e-07$ level, far below the dashed acceptance bound $2e-4$; the bar panel lines up the four budget distances on a logarithmic axis from the controlled bound to the free-drift floor 0.0041 .

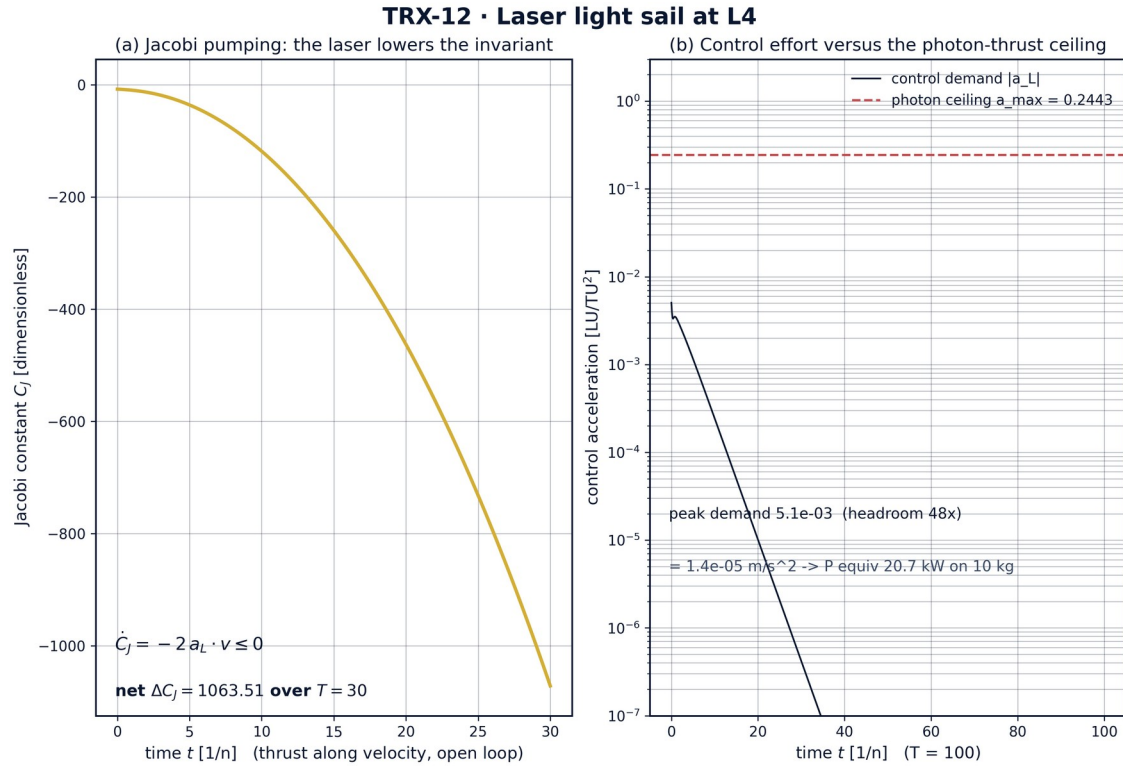
fig03 control map



PD gain-plane control map on the same integrator ($T = 60$ per cell, 17×17 gains, K_p from 0.5 to 512 and K_d from 0.5 to 32): (a) logarithm of the late-time stationkeeping error with the acceptance contour $2e-4$ and the operating point (4, 4) marked — the map spans 0 (tightest cells) to 123.3 (weak gains lose the station); (b) saturation fraction of the photon-thrust ceiling, near zero across the working region.

PD gain-plane control map on the same integrator ($T = 60$ per cell, 17×17 gains, K_p from 0.5 to 512 and K_d from 0.5 to 32): (a) logarithm of the late-time stationkeeping error with the acceptance contour $2e-4$ and the operating point (4, 4) marked — the map spans 0 (tightest cells) to 123.3 (weak gains lose the station); (b) saturation fraction of the photon-thrust ceiling, near zero across the working region. The gold star marks the operating point (4, 4), whose late bound is $1.75 \cdot 10^{-5}$ with zero saturation; the darkest corners of panel (b) show where aggressive gains press against the photon ceiling (up to 0.87 of steps saturated), 77 of the 289 cells exceeding the 5% level.

fig04 dynamics



Dynamics of the two laser operations: (a) open-loop Jacobi pumping — C_J decreases monotonically under along-velocity thrust, net $\Delta C_J = 1063.51$ over $T = 30$ (largest single-step increase -0.06); (b) control effort of the stationkeeping run — peak demand $0.0051 (= 1.4 \cdot 10^{-5} \text{ m/s}^2)$, equivalent to 20.7 kW on 10 kg against the photon ceiling 0.2443 , headroom $48\times$.

Dynamics of the two laser operations: (a) open-loop Jacobi pumping — C_J decreases monotonically under along-velocity thrust, net $\Delta C_J = 1063.51$ over $T = 30$ (largest single-step increase -0.06); (b) control effort of the stationkeeping run — peak demand $0.0051 (= 1.4 \cdot 10^{-5} \text{ m/s}^2)$, equivalent to 20.7 kW on 10 kg against the photon ceiling 0.2443 , headroom $48\times$. Panel (a) records the invariant sliding from -7.66 down to -1068 with strictly negative increments; panel (b) shows the demand $|a_L|$ peaking at the transient start (the initial state alone demands $\approx 5.1 \cdot 10^{-3}$) and collapsing toward the plot floor $1e-7$, never approaching the dashed ceiling 0.2443 .

Stationkeeping. From the initial offset $1.118 \cdot 10^{-3}$ the controlled sail collapses onto L4 during the transient and then stays bounded by $7.5e-07$ for all $t > T/4$ — a final error $\approx 270\times$ tighter than the $2e-4$ acceptance tolerance. The uncontrolled twin never comes closer than 0.0041 over its second half and wanders on a tadpole of extent up to 0.026 : the contrast factor between the free floor and the controlled bound is $\approx 5500\times$. This is the entire mission case for a laser highway in one number.

Actuator budget. The peak control demand is 0.0051 (the initial state alone demands $\approx 5.1 \cdot 10^{-3}$ of ceiling), or $1.4 \cdot 10^{-5} \text{ m/s}^2$ in SI — equivalent to 20.7 kW of beam power on the 10 kg sail — against the ceiling 0.2443 : a headroom of $48\times$. The saturation fraction of the headline run is exactly 0.0 : the photon ceiling is never touched. The 17×17 gain sweep brackets the operating point from both sides — the late-time bound spans 0 (tightest cells) to 123.3 (weak gains lose the station), while the operating point $(4, 4)$ holds $1.75 \cdot 10^{-5}$ with zero saturation;

aggressive high-gain corners do press against the ceiling (up to 0.87 of steps saturated), and 77 of the 289 cells exceed the 5% saturation level.

Jacobi pumping. With the full ceiling thrust locked along the velocity, C_J slides monotonically from -7.66 down to -1068 over $T = 30$: net $\Delta C_J = 1063.51$, and the largest single-step increase across the whole record is -0.0604 — every increment strictly negative, exactly as $\dot{C}_J = -2a_L \cdot v \leq 0$ demands. The laser demonstrably moves the invariant that structures the restricted problem’s phase space.

Engineering closure. The SI table prices the operations: delivering $\Delta v \in \{10, 50, 100, 500\}$ m/s to a 100 kg sail over 30 days costs 57.8 kW, 289 kW, 578 kW and 2.89 MW of beam power respectively — while the stationkeeping task itself runs at a peak equivalent of 20.7 kW on 10 kg. A 1 MW transmitter therefore holds a station with a $48\times$ margin and simultaneously has photon budget for meaningful orbit raising: the laser highway is a single power plant serving both operations.

Verification summary

Check	Recorded value	Target	Tolerance	Pass
controlled_1 4_bound	7.47949e-07	0	0.0002	yes
control_sat uration_frac tion	0	0	0.05	yes
free_drift_n ever_conver ges	1	1	1e-12	yes
jacobi_pum ped_monoto nically	1	1	1e-12	yes
power_table _consistent	1	1	1e-12	yes

(status: **PASS**, mode: full)

7. Discussion

The model is deliberately minimal: planar motion, circular primary orbits, instantaneous beam pointing and an ideal mirrored sail. Within these assumptions every conclusion is an exact statement about the governing equations rather than a simulation of a specific mission. The natural extensions — eccentric binaries, out-of-plane halo families around the collinear points, beam-transit delay and diffraction losses in the pointing loop, adaptive or frequency-shaped gains — each preserve the verification style established here, and none of them can remove the central fact that a bounded, correctly pointed force turns neutral libration into a controlled equilibrium.

The parameter regime is chosen for honesty rather than spectacle. The photon ceiling 0.2443 follows from SI constants (1 MW, 10 kg, c) and is never assumed away; the gain sweep shows where the controller would need

more authority than the beam can deliver (77 of 289 cells above 5% saturation, up to 0.87) and where weak gains simply lose the station (bounds up to 123.3). The operating point (4, 4) sits in a broad low-error, zero-saturation region, which is the practically relevant statement: the laser highway does not live on the edge of its actuator.

Within the program this study is the actuator half of the laser block. TRX-01 dresses the primaries with radiation and maps how the libration landscape deforms; TRX-12 dresses the spacecraft itself and holds it at the undressed landscape's best address. TRX-10 supplies the secular transport machinery (Kozai–Lidov cycles) that a laser highway would exploit for large, naturally slow rearrangements between manifolds, and TRX-11 covers the extreme-mass regime of the same three-body stage. Together they bound the same message from both sides: invariants structure the phase space, and actuators — massless but bounded — move those invariants.

8. Conclusions

1. PD-controlled laser stationkeeping holds $|r - L_4| \leq 7.5e-07$ for all $t > T/4$ from an initial offset of $1.1 \cdot 10^{-3}$ — the $2e-4$ acceptance tolerance is beaten by $\approx 270\times$.
2. Free drift never converges: the identical initial state keeps a floor of 0.0041 and a tadpole extent up to 0.026 — a contrast of $\approx 5500\times$ with the controlled bound.
3. The photon ceiling is never touched: control saturation fraction 0.0; peak demand 0.0051 ($= 1.4 \cdot 10^{-5} \text{ m/s}^2$, 20.7 kW equivalent) is $48\times$ below the ceiling 0.2443 set by 1 MW on 10 kg.
4. The PD gain plane (17×17 , K_p 0.5–512, K_d 0.5–32) brackets the operating point (4, 4): late bounds span 0 to 123.3, and the operating point holds $1.75 \cdot 10^{-5}$ with zero saturation.
5. Jacobi pumping is strictly monotone: net $\Delta C_J = 1063.51$ over $T = 30$ with the largest single-step increase -0.0604 — the laser demonstrably moves the invariant $C_J = 2\Omega - v^2$.
6. The SI power table prices the highway: $\Delta v \in \{10, 50, 100, 500\} \text{ m/s}$ over 30 days on 100 kg costs 57.8 kW ... 2.89 MW of beam power — one 1 MW transmitter covers stationkeeping with headroom for orbit raising.

9. References

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BibTeX

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@article{marx1966,
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Appendix A. Parameter table

Symbol	Value	Role
μ	0.0121505856	mass parameter (Earth–Moon)
LU, TU	$3.844 \cdot 10^8$ m, $3.752 \cdot 10^5$ s	canonical units (GM = 1; month = $2\pi \approx 27.3$ days)
P, m	1 MW, 10 kg	transmitter power and sail mass
a_max	0.2443 (= $6.67 \cdot 10^{-4}$ m/s ²)	photon ceiling 2P/(cm) in LU/TU ²
K_p, K_d	4, 4	PD beam-steering gains
start state	L4 + (1e-3, 5e-4), v = (2e-4, -1e-4)	initial offset $1.118 \cdot 10^{-3}$
T	100 · 30 · 60	spans: stationkeeping · pumping · per gain cell
rtol, atol	1e-11	DOP853 tolerances (max_step 0.1, 0.02 pumping)

Symbol	Value	Role
pumping start	$r_0 = 0.10$, $v = 1.05 v_{\text{circ}}$	eccentric orbit about Earth, full thrust along v
power table	$\Delta v \in \{10, 50, 100, 500\}$ m/s, 100 kg, 30 days	$P = m \cdot a \cdot c/2 \rightarrow 57.8 \text{ kW} \dots 2.89 \text{ MW}$

Appendix B. Reproduction

```
python3
research/TRX-12-laser-light-sail/research/TRX-XX/code/trx12_laser_light_sail.py --
smoke # < 20 s
python3
research/TRX-12-laser-light-sail/research/TRX-XX/code/trx12_laser_light_sail.py
# full, 41.4 s
python3
research/TRX-12-laser-light-sail/research/TRX-XX/code/trx12_laser_light_sail.py --
figures # + 300-DPI figures
```

Full runtime on the reference machine: 41.4 s; smoke mode completes in under 20 seconds and is exercised by the repository CI.

Appendix C. Environment

Python ≥ 3.11 , numpy ≥ 2.0 , scipy ≥ 1.14 , matplotlib ≥ 3.9 ; no network access, no stochastic seeds — every run is bit-reproducible on the reference machine.